

Exercise 4: Employee Management System

Aim

To implement an Employee Management System using arrays in C# and perform basic operations such as add, search, traverse, and delete employee records.

Problem Understanding

Arrays store elements in contiguous memory locations, allowing direct access using an index. They are simple, memory-efficient, and suitable when the number of records is fixed.

Classes Used

Employee

Stores employee information:

- Employee ID
- Name
- Position
- Salary

EmployeeManagement

Implements the following operations using an array:

- Add Employee
- Search Employee
- Traverse Employees
- Delete Employee

Operations Performed

- Add employee records
- Search an employee by Employee ID
- Display all employee records
- Delete an employee record

Time Complexity

Operation Time Complexity

Add $O(1)$

Search $O(n)$

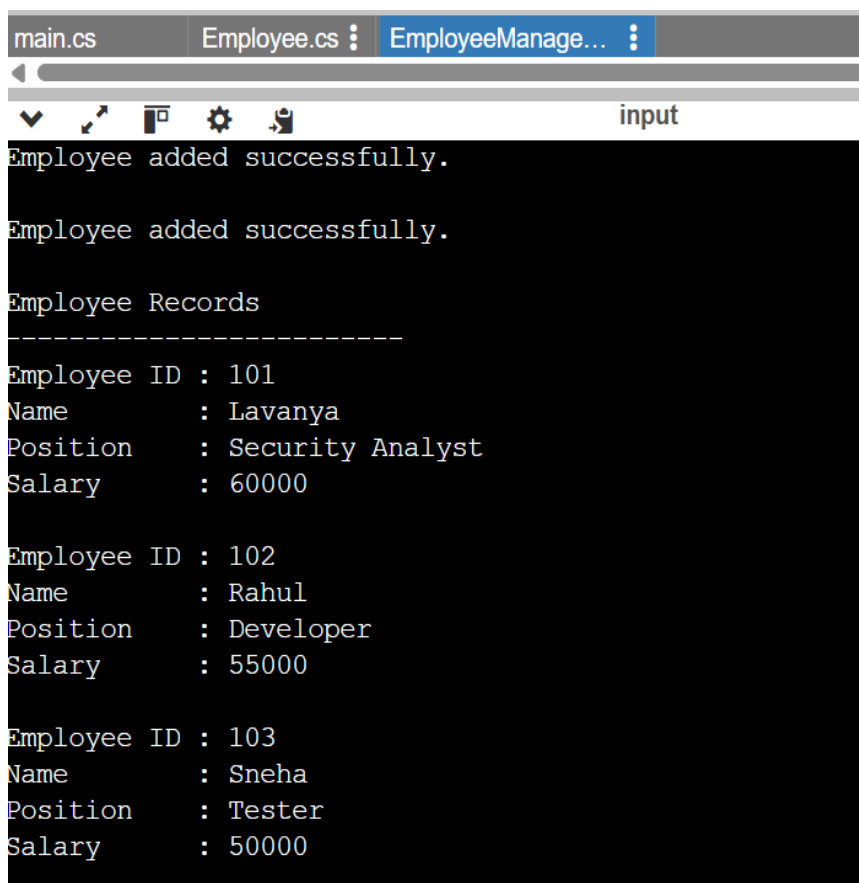
Traverse $O(n)$

Delete $O(n)$

Limitations of Arrays

- Fixed size; cannot grow dynamically.
- Insertion and deletion may require shifting elements.
- Less flexible than collections such as `List<T>`.

Output



```
main.cs  Employee.cs  EmployeeManage...  input
Employee added successfully.
Employee added successfully.
Employee Records
-----
Employee ID : 101
Name       : Lavanya
Position   : Security Analyst
Salary     : 60000

Employee ID : 102
Name       : Rahul
Position   : Developer
Salary     : 55000

Employee ID : 103
Name       : Sneha
Position   : Tester
Salary     : 50000
```

```
input
Employee Found:
Employee ID : 102
Name       : Rahul
Position   : Developer
Salary     : 55000

Employee deleted successfully.

Employee Records
-----
Employee ID : 102
Name       : Rahul
Position   : Developer
Salary     : 55000

Employee ID : 103
Name       : Sneha
Position   : Tester
Salary     : 50000

...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion

This exercise demonstrates how arrays can be used to manage employee records and perform basic CRUD operations. It also highlights the advantages and limitations of arrays in application development.